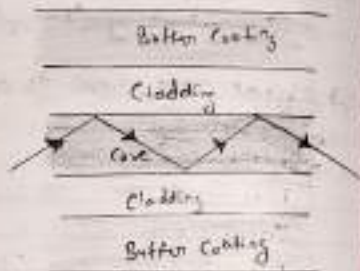
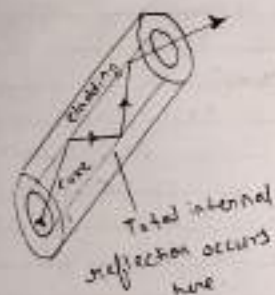


Diagram



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Experiment No. 7

Object:- To measure the Numerical Aperture of an Optical Fibre.

Apparatus:- Optical fibre, Circuit board or a Bread board containing fibre optic analogue transmitters and fibre optic analogue receiver circuits & DC regulated DC power supply.

Theory & formula:- If a light ray is incident on one of the fiber at an angle of θ with the normal, then it follows the snell's law

$$N_1 \sin \theta_1 = N_2 \sin \theta_2$$

where N_1 is the refractive index of the medium air. If N_1 & N_2 are the refractive indices of the material of the core & the cladding then for angle of incidence ' i ' at the cladding layer.

$$i \geq i_c = \sin^{-1}(N_2/N_1)$$

$\sin i_c = N_2/N_1$ and numerical aperture (NA) of the fiber is given by

$$NA = \sqrt{N_1^2 - N_2^2}$$

Calculation

$$NA = \frac{W}{(4L^2 + W^2)^{1/2}}$$

$$(i) NA_1 = \frac{A}{[4(80)^2 + (30)^2]^{1/2}} \quad (ii) NA_2 = \frac{10}{[4(92)^2 + (10)^2]^{1/2}}$$

$$NA_1 = 0.0528$$

$$NA_2 = 0.0643$$

$$\theta_1 = \sin^{-1}(0.0528)$$

$$\theta_1 = 3.01$$

$$\theta_2 = \sin^{-1}(0.0643)$$

$$\theta_2 = 3.72$$

$$(iii) NA_3 = \frac{16}{(4(111)^2 + (20)^2)^{1/2}}$$

$$NA_3 = 0.0659$$

$$\theta_3 = \sin^{-1}(0.0659)$$

$$\theta_3 = 3.79$$

$$(iv) NA_4 = \frac{20}{(4(144)^2 + (20)^2)^{1/2}}$$

$$NA_4 = 0.0692$$

$$\theta_4 = \sin^{-1}(0.0692)$$

$$\theta_4 = 3.96$$

$$(v) NA_5 = \frac{24}{(4(165)^2 + (20)^2)^{1/2}}$$

$$NA_5 = 0.0725$$

$$\theta_5 = \sin^{-1}(0.0725)$$

$$\theta_5 = 4.157$$

$$(vi) NA_6 = \frac{28}{(4(187)^2 + (20)^2)^{1/2}}$$

$$NA_6 = 0.0746$$

$$\theta_6 = \sin^{-1}(0.0746)$$

$$\theta_6 = 4.278$$



$$NA = \frac{W}{(4L^2 + W^2)^{1/2}}$$

Where L is the distance on the IIL (Platform on which observations could be made at different positions) where light emergent from the optical fiber is spread. And W is the spread of the light spot observed on the screen. See fig depicting 1 to W.

Observations:-

S.No	L (mm)	W (mm)	NA	θ (degrees)
1.	6.9	8	0.0528	3.01
2.	9.3	12	0.0543	3.10
3.	12.1	16	0.0659	3.72
4.	14.4	20	0.0692	3.96
5.	16.5	24	0.0725	4.157
6.	18.7	28	0.0746	4.278

Result:- The numerical aperture of the given optical fiber (at 4.000 light) = 0.0673

$$\theta_{\text{mean}} = \frac{3.31 + 3.72 + 3.77 + 3.96 + 4.15 + 4.22}{6}$$

$$\theta_{\text{mean}} = \frac{23.13}{6} = 3.863^\circ$$

At θ_{mean}

$$NA = \sin \theta_{\text{mean}}$$

$$NA = \sin(3.863)$$

$$NA = 0.0673 \text{ Ang}$$

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Precautions:-

1. Attach the fiber optical cord properly
2. Distinguish the outer to inner pink light spots to thus make measurement of 'w'
3. Make sure that wave length selected is 630nm

Industrial Application:-

He-Ne laser is used widely in commercial and industrial applications. It is commonly used in laboratory demonstrations of optics. It is the most common laser used in holography. Other major applications of He-Ne laser include:

- Barcode Scanners
- Tool Alignment
- Non-contact measuring & monitoring
- Blood cell analysis
- Particle counting & fluid seeding
- Alignment of high power CO₂ & YAG treatment lasers & painting beams

- Q5 (b) When mercury light passes through the grating it produces line spectrum while light from laser sources produces spots.
- (c) Laser source higher order maxima can be observed but mercury source has a limit up to second order maxima.

Q5 Is He-Ne LASER a four level energy level laser system?

Ans Yes, He-Ne LASER is a four energy level system.

Q6 Give an example of three level laser.

Ans Ruby LASER.

Q7 Mention the basic characteristics of laser light source.

Ans LASER light is monochromatic, high intense, directional & coherent.

Q8 What is wavelength of light emitted from He-Ne laser?

Ans The wavelength of light emitted by He-Ne laser is $6328 \text{ \AA}$.

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Q8 What is the wavelength of light emitted from He-Ne laser?

Ans The wavelength of light emitted by He-Ne laser is $632.8 \text{ \AA}$.